Practice Problems: AC Power

Introduction to Electric Circuits
Supplemental Instruction

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1 Y Configuration Voltages and Currents

A generator of phase voltage $V_p = 200$ V, in the Y configuration, is connected to a Y-connected load of 20Ω per phase. Find the magnitude of (a) the line voltage, (b) the phase current at the load, and (c) the line current.

Solution

The generator is Y-connected, so $V_l = \sqrt{3}V_p$. Therefore, the line voltage is $V_l = \sqrt{3} \cdot 200\text{V} = 346$ V (a). The phase voltage of the load is the same as the phase voltage of the source, so the phase current can be calculated at the load as $I_p = \frac{200\text{V}}{20\Omega} = 10$ A (b). For Y configuration, the phase and line current are identical, so $I_l = I_p = 10$ A.

2 Y and Δ Configurations

A Y-connected source of $V_l = 150$ V is connected to a delta-configured load of $5 + 10j\Omega$ per phase. Find the magnitude of (a) the phase voltage of the source, (b) the phase current of the source, and (c) the line current.

Solution

For the load, the line and phase voltages are identical, so $V_p = 150$ V (a). The magnitude of the load impedance is $\sqrt{5^2 + 10^2} = 11.18\Omega$, so the load phase current is $I_p = \frac{150\text{V}}{11.18\Omega} = 13.42$ A (b). For the load, $I_l = \sqrt{3}I_p = \sqrt{3} \cdot 13.42\text{A} = 23.24$ A (c).

3 Phase Sequence

A Y-Y system has a source with phase voltage magnitude of $V_p = 50$ V. The per-phase impedance is equivalent to a 5Ω resistance in parallel with a 8Ω capacitive reactance. The phase sequence is BAC, and the angle of the voltage of phase B is zero. Find (a) the phasor voltages of each phase and the phasor diagram, (b) The phasor currents for each phase and the phasor diagram, and (c) the power factor of the load.

Solution

The phase sequence means that B leads A, A leads C, and C leads B. Therefore, $\alpha_B = 0°$, $\alpha_C = 120°$, and $\alpha_A = -120°$. Therefore, the phase voltages are $\boldsymbol{V_A} = 50\angle - 120°$, $\boldsymbol{V_B} = 50\angle 0°$, and $\boldsymbol{V_C} = 50\angle 120°$ (a). For each phase, the impedance is $Z = \frac{1}{\frac{1}{5} - \frac{1}{8j}} = 4.24\angle - 32°\ \Omega$. Therefore, the currents can be found using Ohm's Law: $\boldsymbol{I_A} = \frac{50\angle -120°\ \text{V}}{4.24\angle -32°\ \Omega} = 11.8\angle - 88°$ A, $\boldsymbol{I_B} = \frac{50\angle 0°\ \text{V}}{4.24\angle -32°\ \Omega} = 11.8\angle 32°$ A, and $\boldsymbol{I_C} = \frac{50\angle 120°\ \text{V}}{4.24\angle -32°\ \Omega} = 11.8\angle 152°$ A (b). The power factor can be found as $\cos(\alpha_v - \alpha_i) = \cos(\alpha_z) = \cos(-32°) = 0.848$, leading.

References

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